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ORIGINAL ARTICLE

AI-Powered Water Quality Index Prediction: Unveiling Machine Learning Precision in Hyper-Arid Regions

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Abstract

Water is a vital resource essential for all life, and its quality has been compromised by pollution and contamination in recent decades. The Water Quality Index (WQI) is crucial for evaluating water validity for several purposes, including drinking. Accurate WQI prediction allows for proactive strategies to combat water contamination, preserve public well-being, and guarantee access to safe water sources. This study introduces a novel approach utilizing advanced Machine Learning (ML) techniques for WQI prediction, demonstrating substantial improvements over traditional methods. The methods include the Ridge Model, Lasso Model, Random Forest (RF) Model, Extra Trees (ExT) Model, AdaBoost (AB) Model, XGBoost (XGB) Model, Gradient Boosting (GB) Model, LightGBM Model, Linear Regression (LR) Model, K-nearest neighbor (KNN) Model, Regressor (R) Model, Decision Tree (DT) Model, Multi-layer Perceptron (MLP) Model and Support Vector Regressor (SVR) Model, to determine the most effective models for predicting WQI. The proposed models are trained on a publicly available dataset from 145 groundwater well samples collected between January and April 2018 in Abu Dhabi, the United Arab Emirates (UAE). The models' performance was assessed using various metrics, including Mean Square Error (MSE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Adjusted R-squared, Mean Absolute Percentage Error (MAPE) and R-squared (R2). Experimental results indicate promising performance across all models. In particular, the LR Model proved to be exceptionally accurate, precisely predicting WQI values with 100% accuracy during testing. According to the experimental findings, this model surpassed others in regression tasks, achieving an R^2 value of 100% in WQI prediction. The proposed research confirms the effectiveness of ML algorithms in the field of Water Resources and will serve as a reference for the researchers working in the field of WQI prediction.

Keywords Groundwater Management · Hyper-arid Regions · Machine Learning · Performance Evaluation · Prediction Models · Water Quality Index

1 Introduction

Water resources management is a crucial and systematic process that includes several components such as policy, regulations, institutional framework, sophisticated analytical capabilities, trained labor, well-organized infrastructures, and financial independence. To implement the management policy and restore good water quality status, multiple frameworks are required. The monitoring program is the most applied approach to prioritize water quality assessment. The primary objective is to collect quantitative data on the physical, chemical, and biological characteristics of water using statistical methods. However, due to the specific resource requirements, access to both technical and financial resources is particularly important. Water quality is a critical aspect affecting public health and environmental

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sustainability, particularly in regions facing water scarcity challenges, like the Emirate of Abu Dhabi, UAE (Younis et al. 2023). In today's rapidly changing world, the significance of accurate water quality assessment cannot be exaggerated (Mahmoud et al. 2018). With increasing urbanization, industrialization, and agricultural expansion, ensuring safe and reliable drinking water sources becomes vital for the well-being of the population and the ecosystem (Mishra 2023). A significant challenge to sustainable development in the UAE is the lack of sufficient water resources (Murad et al. 2007). Abu Dhabi's arid environment and limited freshwater resources underscore the significance of groundwater as a primary source of drinking water (Sefelnasr et al. 2023). The quality of groundwater directly influences its suitability for human consumption, highlighting the urgent need for comprehensive assessments to safeguard public health (Ruidas et al. 2023).

The methodological approach of this study builds on the advancements in ML and applies a unique ensemble of multiple models, aiming to outperform traditional water quality assessment techniques. The innovative aspect lies in comparing and optimizing a wide range of ML models for a region-specific application, i.e., the hyper-arid Abu Dhabi area. This multi-model approach enables a more detailed understanding of the Water Quality Index (WQI) predictions and accounts for diverse feature relationships and data limitations specific to hyper-arid environments. To date, numerous tools and techniques have been developed to assess water quality. Among these traditional methods, the WQI stands as a significant model as it allows the conversion of extensive water quality data into simplified numerical values, surpassing the complexity of traditional approaches. This tool has recently gained widespread application in water quality assessment, both in surface and groundwater, owing to its user-friendly mathematical operations. However, this technique has faced criticism, including concerns regarding uncertainty, reliability of the model, transparency, and model sensitivity. Recent studies have highlighted significant uncertainty introduced by the WQI, urging the need for its refinement into a unique approach that accurately determines water quality classification. In response to these limitations, this study not only employs traditional WQI methods but also integrates ML techniques for improved water quality classification and prediction. This approach enhances the model's transparency and reliability, significantly minimizing the uncertainties typically encountered in WQI estimation. Several models such as Long Short-Term Memory (LSTM), Ridge Regression, RF Regressor, KNN, Logistic Regression, XGBoost, Support Vector Machines (SVM), MLP, CATBoost, Lasso, Ridge and Linear Regression have proven effective in predicting WQI and Water Quality Classification (WQC) (Aldris and Farhoud 2020; Dimple et al. 2022; Khan et al. 2022; Khouri & Sallom 2023; Nair & Vijaya, 2022; Vilupuru et al. 2022). LR serves as a foundational approach. Selected for its interpretability, it provides baseline insights into the relationships between water quality parameters and WQI, offering a comparative benchmark for more complex models. These models utilize parameters including pH, temperature, alkalinity, dissolved oxygen, coliform, chloride, hardness, percent sodium (%Na), magnesium hazards (MH), permeability index (PI), sodium adsorption ratio (SAR), Kelly ratio (KR) and soluble sodium percentage (SSP) to accurately predict water quality levels, enabling proactive measures to prevent health risks associated with poor water quality.

Several notable studies have further contributed to this field. For instance, Baig et al. (2024) investigated the effectiveness of advanced ML methods in improving rainfall prediction within hyper-arid environments such as the UAE. Kumar et al. (2023) used LSTM models to predict water table depth in the UAE, showcasing the applicability of ML in environmental predictions. Waraga et al. (2021) explored time series clustering for water consumption patterns in Dubai, UAE, encompassing a crucial aspect of efficient water resource management. Mahmoud et al. (2018) assessed household drinking water quality in Abu Dhabi, UAE, using standard analytical techniques. Osman et al. (2022) summarized the most common methodologies in groundwater level forecasting using artificial intelligence (AI) from studies published between 2010 and 2020. Elmahdy et al. (2021) mapped the groundwater potential on a regional scale using the Classification and Regression Trees (CART) model in the Arabian Peninsula countries, including UAE, Oman, Saudi Arabia, and Yemen. Al-Ruzouq et al. (2019) used weighted overlay analysis and ML to map groundwater potential zones in North UAE, considering factors like geology, precipitation, and geomorphology for sustainable groundwater management. Fang et al. (2024) developed a comprehensive and affordable isotopic database that effectively identifies the recharge sources for aquifers across the UAE. This initiative not only enhances our understanding of water resources but also supports sustainable management practices for our vital water supply. Al-Ruzouq et al. (2023, 2024) utilized ML techniques with Geographic Information System (GIS) and remote sensing images to construct a high-efficiency Artificial Groundwater Recharge (AGR) map for the United Arab Emirates. Elmahdy et al. (2020) measured flash flood magnitude in the UAE while integrating ML and geohydrological models. (Wee et al. 2021) reviewed the models for water level forecasting in reservoirs based on Yagoub et al. (2022) created an urban quality of life index map using remote sensing and ML for Al Ain, UAE. Elhaj et al. (2023) explored the use of Artificial Neural Network technology in Al Ain, UAE, to forecast

fluctuations in groundwater levels. Other notable studies include reviewing groundwater quality in Algeria using hydrochemical parameters and AI techniques (Derdour et al. 2023) and evaluating groundwater quality for irrigation in Sohag, Egypt, utilizing Deep Learning Time Series Techniques (Ahmed et al. 2023). Ibrahim et al. (2023) assessed groundwater quality for irrigation purposes in Egypt using WQIs, GIS, and ML models, showcasing accurate prediction of groundwater quality indices in arid environments. Sahour et al. (2022) identified shallow groundwater occurrences in arid lands of the Western Desert of Egypt using multi-sensor remote sensing data and ML algorithms. SVM and XGB were used to forecast groundwater aflaj potential in the Sultanate of Oman (Al-Kindi and Janizadeh 2022). Hanoon et al. (2022) developed an approach based on public participation and GIS coupled with ML in the water scarcity risk prediction at Nasiriyah, Iraq. Mohammed et al. (2023) assessed the suitability of groundwater by neural network and support vector regression (SVR) in the northern Khartoum area, Sudan.

The current study attempts to bridge the gap between traditional techniques for water quality assessment and innovative ML approaches. The authors aspire to advance the field of water quality monitoring and promote environmental resilience in arid regions. The main objectives are to detect the most effective and dependable ML algorithms to forecast WQIs and conduct a thorough evaluation of groundwater quality in the hyper-arid area of Abu Dhabi, UAE, applying advanced ML models. Additionally, the objectives include providing a comparative analysis for predicting the WQI and analyzing and interpreting the research findings to provide insights into the benefits and challenges associated with incorporating ML techniques in groundwater quality evaluation.

There has been an increased concern regarding groundwater contamination in the Emirate of Abu Dhabi during the past few years, which poses significant challenges to water quality management (Baig et al. 2023; Sherif et al. 2023). Addressing these challenges requires a thorough understanding of the physio-chemical parameters and their impact on water quality, particularly concerning drinking water standards. Similarly, ensuring the management and long-term usage of this valued resource depends on understanding and analyzing water quality (Shams et al. 2023).

Building upon the developments in water quality assessment and management, several notable studies have contributed to the field.Hussein et al. (2023) explored various ML algorithms for predicting the WQI, whileDharma et al. (2023) reviewed a wide range of ML methods for water quality prediction.Tabassum et al. (2023) conducted a performance analysis of different ML techniques using physiochemical parameters, and Alshehri et al. (2023) coupled machine and deep learning with explainable AI to enhance groundwater quality prediction and decision-making in Saudi Arabia. The case study of Goodarzi et al. (2023) in Iran, demonstrated the application of ML algorithms for WQI estimations, and Wang et al. (2023) developed comprehensive weighting methods integrated with ML for water quality prediction.Miller et al. (2023) advanced water quality monitoring through artificial neural networks, presenting current insights and future research opportunities, andWilliam et al. (2023) has created AI-based models to support water quality prediction using an ML approach. This study explores the use of advanced ML models for predicting the WQI and identifying the most reliable methods. By comparing various ML algorithms, the research evaluates their effectiveness in supporting groundwater quality management, particularly in the hyper-arid region of Abu Dhabi, UAE. The findings will offer insights into the potential of ML techniques to enhance environmental monitoring and water resource management.

2 Materials and methods

The proposed study comprised three primary modules, as depicted in Fig. 1. The initial phase was dedicated to mapping the application domain and data acquisition. Following this, the second phase involved implementing models and optimizing their parameters. Lastly, the third phase involved evaluating the prediction results using a range of evaluation metrics.

2.1 Application Domain: Abu Dhabi Area

Abu Dhabi, the capital of the UAE, is situated on the southeastern coast of the Arabian Peninsula along the Arabian Gulf. It is bounded to the east by Oman, to the south and west by Saudi Arabia, and to the northeast by Dubai. As the largest of the seven emirates that make up the UAE, Abu Dhabi encompasses an area of 67,340 square kilometers, covering about 87% of the nation's total land area. According to the Abu Dhabi Statistics Centre, around mid-2016, the estimated population of the emirate was approximately 3 million, marking it as the highest population in the UAE. A significant portion of this population consists of expatriates, contributing to its status as one of the fastest-growing urban centers in the world. Abu Dhabi has a low-lying coastline (Fig.2a) extending for more than 400 km comprising around 200 islands with variable sizes.

The climate of Abu Dhabi is classified as hyper-arid, characterized by extremely high temperatures, especially during the summer months, and minimal annual rainfall. The average temperatures range from 42 °C in the summer

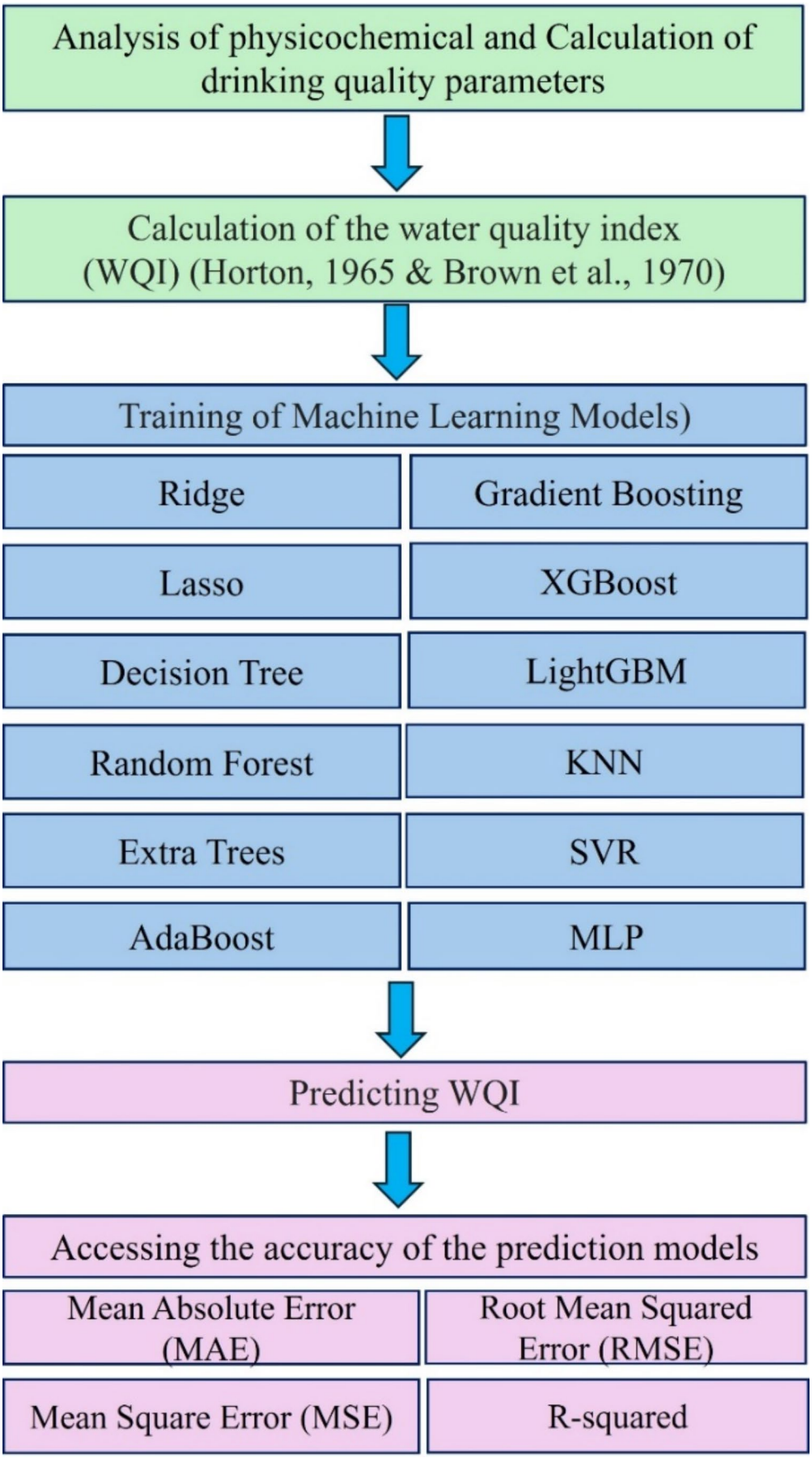


Fig. 1 Flow chart architecture of the methodology adopted to predict WQI

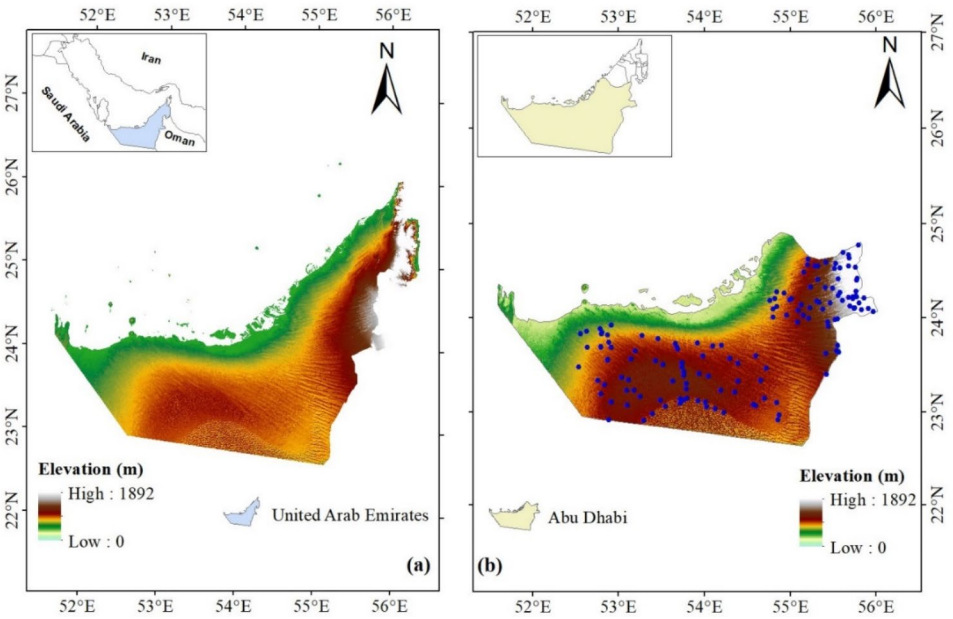


Fig. 2 (**a**) Digital Elevation Model of UAE and (**b**) Map showing the location of the study area and investigated wells

to 13 °C in the winter, with peak temperatures often exceeding 45 °C. The region experiences very low annual precipitation, averaging less than 100 millimetres, typically occurring in short, sporadic bursts.

Abu Dhabi faces significant challenges regarding water resources, primarily due to its arid climate and limited natural freshwater sources. The emirate relies heavily on desalination processes to meet the water demands of its population and industries. Groundwater, which accounts for a substantial portion of the non-desalinated water supply, is mostly brackish and overexploited, leading to concerns over sustainability and quality. Freshwater resources are limited and are mainly utilized for agricultural purposes, further stressing the already limited supplies. The scarcity of natural freshwater resources in Abu Dhabi is a critical issue worsened by the region's high rate of urbanization, population growth, and economic development. Additionally, groundwater depletion has led to increased salinity and contamination, reducing the availability of drinking water. Addressing these challenges requires comprehensive water management strategies, focusing on sustainable desalination technologies, efficient water use practices, and the protection and recharge of groundwater resources. Assessing water quality was essential for planning sustainable solutions in Abu Dhabi, where the proposed framework was utilized to evaluate water quality effectively.

2.1.1 Groundwater Salinity Challenges

Groundwater salinity is a major ecological concern in the Arabian Gulf, presenting significant threats to urban life and agricultural production in the region (Alhamd and Ibrahim 2024). In Abu Dhabi, this issue further complicates sustainable groundwater management, underscoring the importance of developing a comprehensive and context-specific framework for sustainable groundwater management in such arid regions. The Groundwater Research Project – GWRP (conducted between 1988 and 1997) estimated a total groundwater reserve of 7% fresh and 93% brackish, while the Groundwater Assessment Project- GWAP (GTZ et al., 2005) calculated a total estimate of 79.4% saline (Fig. 3).

2.2 Data Acquisition, Statistics and Correlation

In the proposed work, a publicly available dataset (Batarseh et al. 2021) was utilized to predict the groundwater quality for drinking purposes in arid regions. The data was collected by the Environment Agency Abu Dhabi (EAD) focused on investigating groundwater quality in specific hydrogeological areas within the Abu Dhabi Emirate. Two primary regions, Al Ain and Al Dhafra, were selected for the study, comprising 39 and 60 grid cells, respectively. A total of 145 water samples were gathered from groundwater wells within Abu Dhabi (as shown in Fig. 2b). The dataset consisted of 145 instances and physicochemical parameters, including

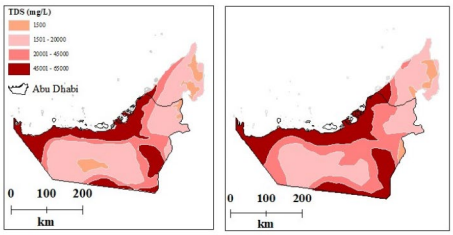


Fig. 3 Groundwater salinity with minor occurrences of fresh water (<1500 mg/L TDS) and mainly brackish-saline-hypersaline groundwater resources above this value in Abu Dhabi and the rest of the UAE

Table 1 Summary statistics of the physicochemical parameters and the calculated WQI for the studied samples

	Number	Mean	Standard deviation	Minimum	Maximum
pH	145	7.47	0.43	6.76	9.08
EC	145	14,120	10,956	968	62,000
K+	145	54.22	41.3	5.27	213
Mg +	145	259	231.9	6	1565
WQI	145	304.52	217.62	62.98	1364.83

pH, EC, $Mg^{2+,}$ and K^{+} as features. The pH of water indicates its acidity or alkalinity level.

Statistical computations on the dataset attributes are provided in Table 1, incorporating parameters such as minimum, maximum, mean and standard deviation values. These metrics present an understanding of the distribution and characteristics of the data. Additionally, the correlation matrix of the dataset features was examined (Fig. 4), revealing relationships between variables and highlighting significant associations or dependencies among them.

2.3 WQI Calculation and Interpretation

The WQI serves as a key indicator influencing water quality, consolidating extensive datasets into a single numerical representation of the overall water quality level (Brown et al. 1970; Saeedi et al. 2010). It serves as a vital tool for evaluating the overall groundwater quality, integrating multiple water quality parameters into a single numerical value. Developed by (Horton 1965) to provide a concise and objective evaluation, the WQI facilitates the interpretation and communication of complex water quality data to policymakers, researchers, and the public.

WQI is computed by utilizing different parameters while using Eq. 1.

$$\mathrm{WQI} = \frac{\sum_{i=1}^{N} q_i \times w_i}{\sum_{i=1}^{N} w_i} \quad (1)$$

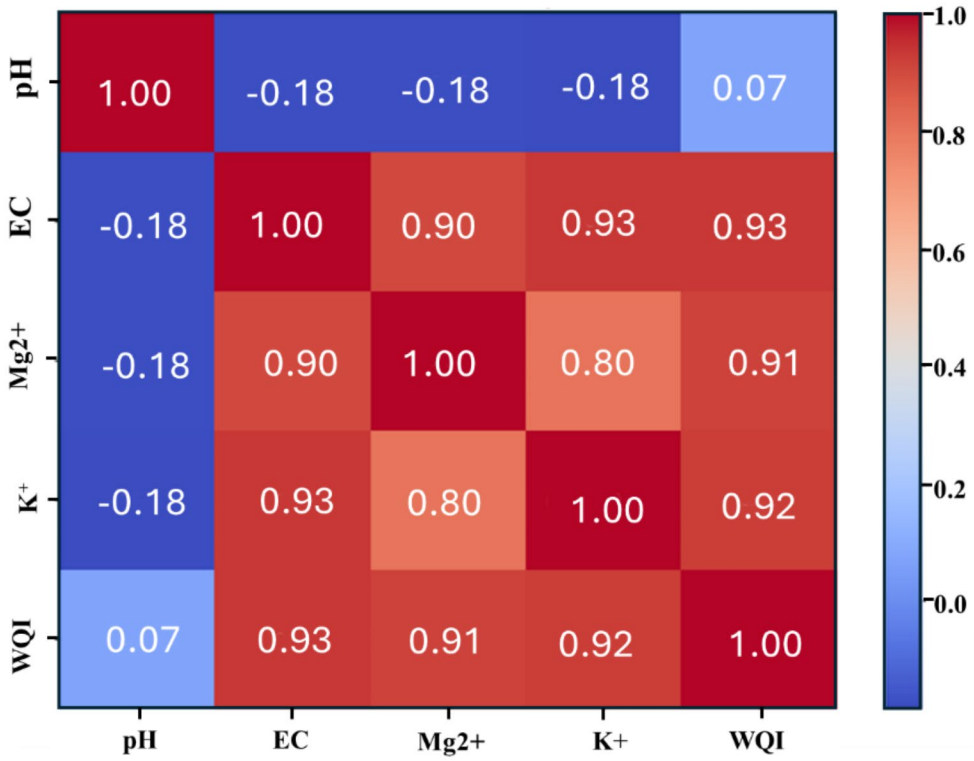


Fig. 4 Correlation Heatmap showing relationships among physicochemical parameters in groundwater samples

where N denotes the overall parameters, w_i indicates the unit weight for parameter i, and q_i signifies the quality rating scale for parameter i. The q_i is calculated utilizing Eq. 2.

$$q_i = 100 \times \left(\frac{v_i - v_{id}}{s_i - v_{id}} \right) \tag{2}$$

where v_i and v_{id} represent the measured and ideal values for the parameter i, respectively, under pure water conditions, and s_i signifies the standard value for the parameter i. Equation 3 is used to compute the unit weight w_i.

$$w_i = \frac{k}{s_i} \tag{3}$$

where k is the constant of proportionality, which can be determined by applying Eq. 4.

$$k = \frac{1}{\sum_{i=1}^{N} s_i} \tag{4}$$

2.3.1 Classification of the WQI

The WQI varies based on the parameters and methods used (Ahmad et al. 2015) to evaluate water's suitability for drinking purposes and this assessment involves evaluating various parameters on a standardized scale (Kumari and Rai 2020). The WQI quantifies water quality, with lower values indicating better quality (Ahmad et al. 2024). WQI is essential for simplifying and communicating water quality information (Akhtar et al. 2021; Banda and Kumarasamy 2020; Ewaid et al. 2020; Mogane et al. 2023; Suwadi et al. 2022). It consolidates multiple water quality parameters into a single, easily interpretable numerical score (Adimalla 2021; Uddin et al. 2023), which is vital for monitoring public health, ecosystem stability, and water resource management (Li et al. 2022; Lukhabi et al. 2023; Nong et al. 2020). Water quality is classified by the WQI into the following categories: excellent (WQI < 25), good ($25 \leq$ WQI < 50), fair ($50 \leq$ WQI < 75), poor ($75 \leq$ WQI < 100), and very poor (WQI $\geq$ 100) (Wu et al. 2021; Xia et al. 2022; Yang et al. 2023). It serves a critical role in public health and environmental policy by providing clear, actionable insights into water quality (Gao et al. 2020; Jibrin et al. 2024; Li and Wu 2019). This index supports decision-making for resource allocation, water treatment interventions, and regulatory compliance, especially in regions facing water scarcity (Drogkoula et al. 2023; Farzana et al. 2024). By converting complex data into accessible categories, WQI enables proactive measures to safeguard communities and ecosystems.

2.4 Machine Learning Models

In the proposed work, the acquired data is given to various ML models for training purposes. Each model is explained in detail below;

2.4.1 Linear Regression (LR)

In this study, a variety of machine learning models were employed to predict the WQI, with LR serving as a foundational model. It is a statistical approach used to model the relationship between independent variables (such as pH, EC, Mg^{2+}, and K^{+}) and the dependent variable, WQI. As a baseline model, LR helps illustrate the influence of each water quality parameter on WQI, providing insights into their relative contributions. The effectiveness of LR in water quality prediction has been demonstrated in studies by Aldris et al. (2020) and Khouri et al. (2023), highlighting its reliability as a predictive model in environmental assessments. In the context of predicting the WQI, linear regression can be used to establish a mathematical equation (i.e., the best-fitting straight line) that relates the WQI to various water quality metrics, like pH, EC, $Mg^{2+,}$ and K^{+}. This line is represented by Eq. 5.

$$\mathrm{WQI} = a + b_1(x_1) + b_2(x_2) + \ldots + b_n(x_n) \tag{5}$$

where WQI signifies the predicted water quality, a represents the intercept and $b_1, b_2 \ldots, b_n$ coefficients that depict the variation in WQI values for a unit change in the corresponding independent variables ($x_1, x_{2\ldots,}x_n$). The water quality parameter is represented by $x_1, x_{2\ldots,}x_n$.

2.4.2 Ridge

Ridge regression is a technique used to address high correlation between independent variables in linear regression models. When predicting the WQI, there are multiple water quality parameters (e.g., pH, EC, $Mg^{2+,}$ and K^{+}) used as independent variables, and some of these variables could be correlated. It handles the shortcoming of the LR model in dealing with highly correlated data using penalized least squares to avoid a sparse model and can be calculated by squaring the magnitudes of the coefficients (Ogutu et al., 2012). The ridge regression coefficient can be calculated as Eq. 6;

$$\hat{\beta}_{\mathrm{Ridge}} = \mathrm{argmin} \ \| Y - X\beta \|_2^2 + \lambda \| \beta \|_2^2 \tag{6}$$

where Y, X and $\lambda > 0$ represents the dependent, independent, and tuning parameters respectively.

2.4.3 Lasso Regression

Lasso regression is known for its effectiveness in handling datasets with numerous variables, particularly in scenarios where systematic and efficient methods are essential. However, its stability diminishes when dealing with highly correlated predictors. Unlike ridge regression, Lasso tends to encourage sparsity in the model, leading to a greater number of coefficients being pushed toward zero while retaining only a select few with larger magnitudes. The estimator is defined by Eq. 7 as indicated.

$$\hat{\beta}_{\text{lasso}} = \text{argmin} \ \| \mathrm{Y} - \mathrm{X}\beta \ \|_2^2 + \lambda \| \ \beta \ \|_1 \tag{7}$$

where Y, X and $\lambda > 0$ represents the dependent, independent and tuning parameters respectively.

2.4.4 Decision Tree (DT)

The DT model is constructed using the provided water quality dataset. Through the DT algorithm, an iterative process identifies the optimal tree structure by minimizing a fitness function. Each independent water quality factor is utilized to fit the output WQI value. The dataset undergoes partitioning based on different splitting points of these independent features. At each split point, the algorithm calculates the error between the actual and predicted values using a predefined fitness function. This process recurs until the optimal decision tree structure is achieved.

2.4.5 Random Forest (RF)

Similar to the DT algorithm, the RF also generates multiple decision trees instead of just one. These trees collectively aid in determining the best tree to estimate the WQI value. In this process, the water quality factors act as features, and alongside the WQI values, they contribute to the creation of several decision trees.

Employing RF regressor method for predicting the WQI value enables us to attain an impartial assessment of the trees' generalization error. The utilization of Gini impurity aids in determining the probability of misclassification at each node. One notable advantage of this algorithm lies in its ability to maintain satisfactory regression accuracy even when dealing with diminutive or incompletely available datasets. Nonetheless, there's a risk of overfitting the data and introducing noise in regression tasks.

2.4.6 Extra Tree (ExT)

The ExT serves as an extension of RF and functions as an ensemble of DT. Unlike RF, ExT utilizes the entire training set to train all the trees and generates predictions by averaging the outputs from the individual decision trees. This method is characterized by high randomization, making it less susceptible to overfitting in comparison to RF.

2.4.7 AdaBoost

AdaBoost Regressor can significantly enhance the prediction of WQI by iteratively refining its models based on the specific challenges encountered in water quality prediction. Initially, it trains a weak learner, typically a DT, on the dataset, attempting to predict WQI values accurately. In subsequent iterations, AdaBoost focuses more on the instances where the weak learner struggled, assigning them higher weights to prioritize correct prediction. By continually adjusting and combining multiple weak learners, AdaBoost effectively captures the intricate relationships between various water quality parameters and the resultant WQI. This adaptability enables it to address the complexities inherent in WQI prediction, leading to more accurate and robust models for assessing water quality levels. The final prediction $f(x)$ of the model is computed as a weighted sum of weak learners' predictions as shown in Eq. 8 below.

$$\mathrm{f}(\mathrm{x}) = \sum_{\mathrm{t}=1}^{\mathrm{T}} \alpha_{\mathrm{t}} . \mathrm{h}_{\mathrm{t}}(\mathrm{x}) \tag{8}$$

$h_t(x)$ represents the weak learner trained on the weighted dataset and α_t represent the weight of the weak learner.

2.4.8 Gradient Boosting (GB)

GB optimizes a loss function by sequentially adding weak learners to minimize the error between actual and predicted WQI values. Unlike AdaBoost Regressor, which assigns weights to training instances, GB fits each weak learner (usually DT) to the residuals of the previous learner. The final prediction $f(x)$ of the model is computed as a weighted sum of weak learners' predictions as shown in Eq. 9 below.

$$\mathrm{f}(\mathrm{x}) = \sum_{\mathrm{t}=1}^{\mathrm{T}} \eta \, . \mathrm{h}_{\mathrm{t}}(\mathrm{x}) \tag{9}$$

At each iteration t, fits a weak learner $h_t(x)$ to the negative gradient of the loss function about the predictions of the preceding model. To gradually reduce the error, the weak learner is added to the model and weighted by a learning rate η.

2.4.9 XGBoost (XGB)

XGB enhances gradient boosting by incorporating several innovations, including regularization techniques and parallel computing, to achieve exceptional predictive power. It operates similarly to GB, iteratively adding weak learners to minimize the loss function. However, XGB introduces several enhancements such as regularization for overfitting prevention, tree pruning for minimal loss reduction and reducing complexity, parallel computing for accelerating training speed, and customizable loss functions for flexibility in model optimization.

2.4.10 10 Light Gradient Boosting (LightGBM)

LightGBM, is another strong GB framework for predictive modeling, known for its efficiency, speed, and scalability. LightGBM minimizes a differentiable loss function that evaluates the discrepancy between actual and predicted values for WQI. LightGBM builds DT in a leaf-wise manner rather than level-wise, selecting the leaf with the maximum loss reduction to split at each node. This approach can lead to faster training times and better accuracy, notably for large datasets. LightGBM optimizes tree parameters and regularization terms using gradient-based methods, ensuring efficient training and improved performance.

2.4.11 11 K-Nearest Neighbors (KNN)

KNN is a simple yet very efficient algorithm for predicting WQI. In the training phase, KNN stores the features and corresponding WQI values of available water quality samples. When predicting the WQI for a new sample, KNN finds the nearest neighbors established on a selected distance measure, then calculates their weighted average or applies majority voting. Crucial to its performance is the careful selection of k and the distance metric, typically determined through techniques like cross-validation. KNN offers a simple and intuitive solution for WQI prediction, particularly suited for smaller datasets, but parameter selection and normalization are essential for optimal results.

2.4.12 12 Support Vector Regression (SVR)

SVR seeks to maximize the margin by identifying the hyperplane that best fits the training set and is based on the ideas of SVM. Following the identification of the ideal hyperplane, SVR predicts the WQI value $\widehat{y}$ for a new sample using the decision function in Eq. 10:

$$\widehat{\mathrm{y}} = \mathrm{w}^{\mathrm{T}}\varphi\ (\mathrm{x}) + \mathrm{b} \tag{10}$$

In summary, SVR for WQI prediction involves finding the optimal hyperplane that best matches the training data while reducing the margin of tolerance. Through the kernel trick, SVR can handle nonlinear relationships in the data, providing a flexible and effective solution for water quality assessment.

2.4.13 13 Multilayer Perceptron (MLP)

MLP presents a potent neural network model for WQI prediction. It comprises input, hidden, and output layers, with neurons interconnected by adjustable weights. During training, backpropagation adjusts weights to minimize prediction error using nonlinear activation functions like ReLU. Loss functions such as mean squared error quantify prediction accuracy, while regularization techniques like dropout prevent overfitting. Tuning hyperparameters, including the number of layers and neurons, is crucial for optimal performance. Once trained, MLP predicts WQI values by passing features through the network. In essence, MLP's effectiveness lies in its ability to learn complex patterns in water quality data, offering accurate predictions through iterative weight adjustments.

2.5 Evaluation Metrics

Several evaluation metrics were employed to evaluate how well the LR model predicted the WQI. These metrics offer a quantitative measurement of how well the model's predictions match the actual values that were observed. The evaluation metrics used in this study are:

The coefficient of Determination (R-squared) is a statistical evaluation that displays the proportion of the variance in the dependent variable (WQI) that is indicated by the independent variables (parameters related to water quality). Higher numbers indicate a better fit of the model to the data, and it spans from 0 to 1. It is mathematically represented in Eq. 11 below.

$$\mathrm{R}^2 = 1 - \frac{\text{sum of squared residuals}}{\text{Total sum of squares}} \tag{11}$$

MAE quantifies the average size of prediction errors in a set without taking direction into account. Equation 12 shows its computation as the mean of the absolute variances between the values that are observed and those that are predicted.

$$\mathrm{MAE} = \left(\frac{1}{\mathrm{N}}\right) \sum |\mathrm{y_i} - \widehat{\mathrm{y}}_\mathrm{i}| \tag{12}$$

where N is the total number of observations, while y_i and $\widehat{y_i}$ are the observed and predicted values, respectively.

MSE, calculated as the average of the squared differences between observed and predicted values, measures the average squared difference between these values, as depicted in Eq. 13.

$$\mathrm{MSE} = \left(\frac{1}{\mathrm{N}}\right) \sum |\mathrm{y_i} - \widehat{\mathrm{y}}_\mathrm{i}|^2 \tag{13}$$

RMSE is computed by taking the square root of the mean of the squared differences between the observed and projected values and computes the mean magnitude of errors in a set of predictions by assigning greater weight to more significant errors. The formula is given below in Eq. 14.

$$\mathrm{RMSE} = \mathrm{sqrt}\left(\left(\frac{1}{\mathrm{N}}\right) \sum |\mathrm{y_i} - \widehat{\mathrm{y}}_\mathrm{i}|^2\right) \tag{14}$$

3 Results

3.1 Descriptive Statistics and Correlation Analysis

Table 1 includes data on pH, EC, concentrations of K^+ and Mg^+, as well as the overall WQI values derived from these parameters. Across 145 samples, pH exhibited a mean of 7.47 with a standard deviation of 0.43, ranging from 6.76 to 9.08. EC ranged widely from 968 to 62,000 µS/cm, with a mean of 14,120 µS/cm and a standard deviation of 10,956 µS/cm. K^+ and Mg^+ concentrations displayed means of 54.22 mg/L and 259 mg/L, respectively, with varying standard deviations and ranges. The calculated WQI values ranged from 62.98 to 1364.83, with an average of 304.52 and a standard deviation of 217.62. This comprehensive dataset provided a detailed overview of the physicochemical characteristics and overall water quality assessment for the samples under study.

Figure 4 displays a correlation heatmap illustrating the relationships among key physicochemical parameters and the WQI in groundwater samples. The heatmap visually represents Pearson correlation coefficients ranging from −0.18 to 1.00, indicating the strength and direction of correlations between pairs of variables. Specifically, pH shows weak to negligible correlations with EC ($r = -0.18$), Mg ($r = -0.18$), K ($r = -0.18$), and WQI ($r = 0.07$). EC exhibits strong positive correlations with Mg ($r = 0.90$), K ($r = 0.93$), and WQI ($r = 0.93$), highlighting their interdependencies. Similarly, Mg and K demonstrate notable positive correlations ($r = 0.80$), while WQI shows moderate positive correlations with EC ($r = 0.93$), Mg ($r = 0.91$), and K ($r = 0.92$). This graphical representation supports understanding the associations among these parameters, providing insights into their collective influence on groundwater quality assessment.

3.2 Assessment of the Significance of Input Features

The experimental study aimed to develop and evaluate state of the art model for predicting the WQI based on various water quality parameters. The experiments were carried out utilizing Python programming on a desktop system equipped with Alienware Aurora R8. The system features an Intel Core i9-9900k CPU running at 3.60 GHz, 32 GB of RAM, and is powered by an NVIDIA GeForce RTX 2080 GPU. Feature importance was performed to assess the relative significance of input features in a ML model's prediction. For tree-based models, such as DT, RF, ExT, GB, XGB, and LightGBM, feature importance can be directly calculated by measuring how frequently each feature contributes to the model's performance. This is typically done by evaluating how often a feature is utilized to make decisions across all the trees in the ensemble. However, for linear models like LR, Ridge Regression, and Lasso Regression, the concept of feature importance is not as straightforward. In these models, each feature's coefficient represents the direction and strength of its relationship with the target variable. A large coefficient (in absolute terms) indicates that the feature has a strong influence on the predicted outcome, while a coefficient close to zero implies that the feature has little to no influence. The results of the feature importance of all the ML models are shown in Fig. 5 where the features importance is different in the case of different models.

To gain a deeper understanding of the data and its characteristics, distribution plots and box plots (Fig. 6) were generated for all the features involved in predicting the WQI. The distribution plot of the features is shown in Fig. 6, which is a normal distribution and does not exhibit skewness or other deviations from normality. Moreover, the box plots of each feature signify a concise representation of the distribution of each feature, highlighting the median, interquartile range, and the presence of any outliers.

3.3 Performance of the Applied Algorithms

The experimental results of all the ML models used in this study are presented in Table 2. In the experiments, 5-fold cross-validation is employed to evaluate the proposed models and generate substantial and reliable results. The dataset is split into five equal parts, with one validation set and the remaining portions being used as training data. The performance metrics (MAE, MSE, RMSE, R^2 and Adjusted R^2) are averaged across each iteration. This ensures that all data points are used for both training and testing. The evaluation

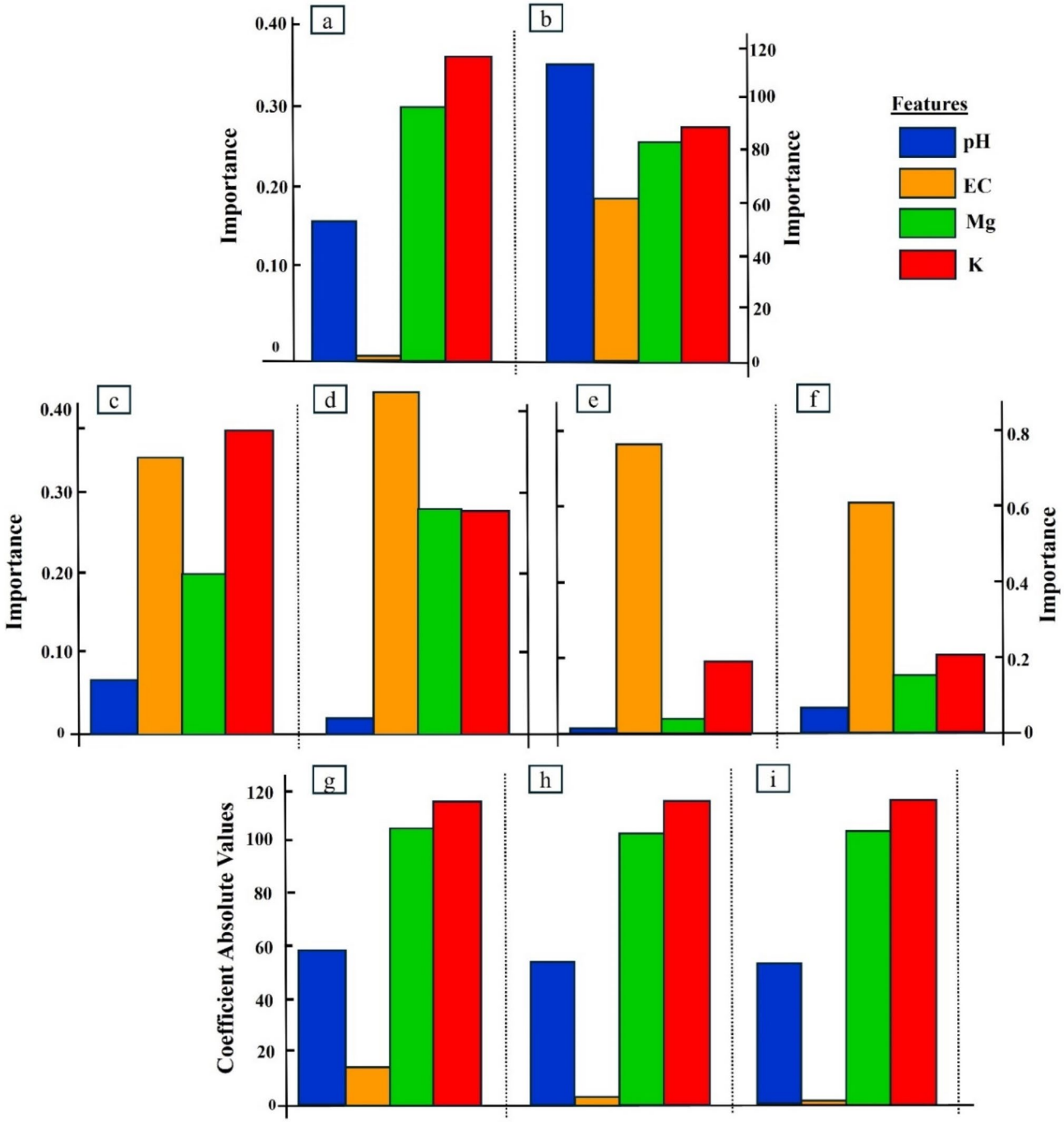


Fig. 5 Comparison of Feature Importance/Coefficients across different ML Models including a) GB Model, b) LightGBM Model, c) ExT Model, d) XGB Model, e) DT Model, f) RF Model, g) Ridge Model, h) Lasso Model and i) LR Model

of the models for predicting the WQI revealed distinct performances across various techniques. LR and Lasso Regression exhibited remarkable accuracy, with LR attaining perfect scores (MAE = 0.00, MSE = 0.00, RMSE = 0.00, R-Square = 1.00), indicating a remarkably precise linear relationship between the WQI and the water quality parameters. Lasso Regression also performed extremely well, with an R-Square of 1.00, an MSE of 0.64, an MAE of 0.70, and an RMSE of 0.80. Ridge Regression, while not as precise, still demonstrated a robust linear relationship and an overall good fit, as evidenced by its reasonably low error values (MAE = 2.22, MSE = 10.00, RMSE = 3.16) and a high R-Square score of 0.99.

Among the tree-based models, ExT and GB emerged as the top performers, boasting the lowest error values (MAE = 10.66, MSE = 206.09, RMSE = 14.36 for ExT

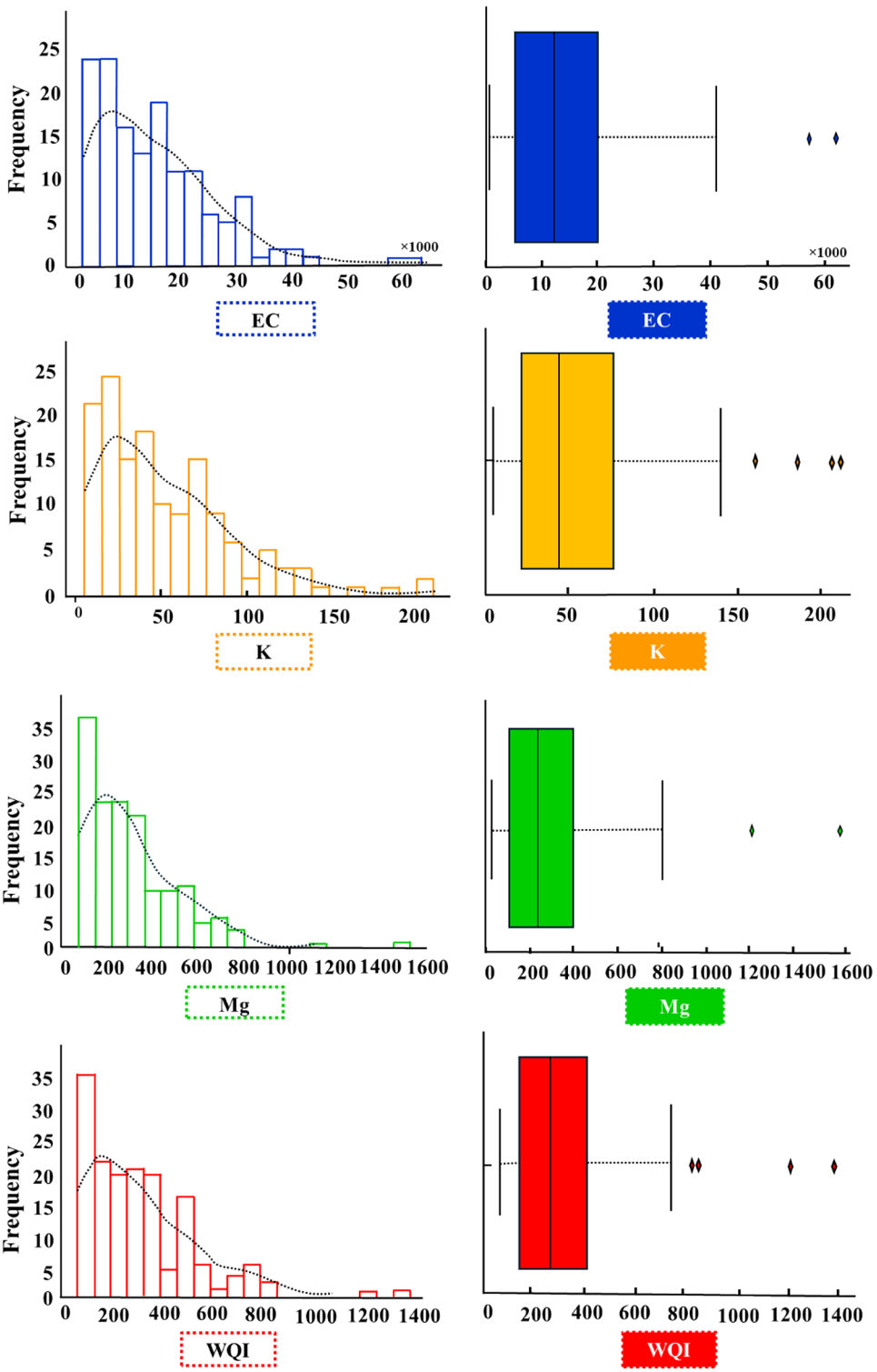


Fig. 6 Distribution of Electrical Conductivity, Potassium, Magnesium, and calculated WQI values along with their corresponding Box plots

and MAE = 12.51, MSE = 242.39, RMSE = 15.56 for GB) and highest R-Square scores of 0.99. These models excelled in capturing the non-linear relationships between the water quality parameters and the WQI, resulting in highly accurate predictions. RF also showcased its ability to model complex relationships and handle diverse data characteristics effectively, achieving a high R-Square score of 0.97 (MAE = 16.91, MSE = 533.74, RMSE = 23.10). However, AdaBoost and LightGBM exhibited relatively higher error values (MAE = 41.02, MSE = 2329.58, RMSE = 48.27 for AdaBoost and MAE = 33.37, MSE = 3021.08, RMSE = 54.96 for LightGBM) and lower R-Square scores

Table 2 Performance of the regression models

Models	Evaluation Metrics				
	MAE	MSE	RMSE	*R*-Square	Adjusted *R*-Square
LR	0.0000	0.0000	0.0000	1.0000	1.0000
Lasso	0.6967	0.6438	0.8024	1.0000	1.0000
Ridge	2.2183	9.9962	3.1617	0.9996	0.9995
DT	19.6904	862.0754	29.3611	0.9625	0.9562
RF	16.9174	533.7494	23.1030	0.9768	0.9729
ExT	10.6640	206.0891	14.3558	0.9910	0.9895
AdaBoost	41.0230	2329.5771	48.2657	0.88986	0.8817
GB	12.5056	242.3984	15.5691	0.9894	0.9877
XGB	16.5974	600.8892	24.5130	0.9738	0.9695
LightGBM	33.3745	3021.0760	54.9643	0.8685	0.8466
KNN	13.2239	289.5283	17.0155	0.9874	0.9853
SVR	119.0946	17001.7185	130.3906	0.2599	0.1366
MLP	104.0320	14865.8315	121.9255	0.3529	0.2450
Ensemble 1	8.945	148.3303	12.1791	0.9935	0.9925
Ensemble 2	16.5801	594.8799	24.3902	0.9741	0.9698
Ensemble 3	7.2881	107.4721	10.3669	0.9953	0.9945

Ensemble 1: Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting, and XGBoost. Ensemble 2: Linear Regression, Random Forest, XGBoost, and LightGBM. Ensemble 3: Ridge, Lasso, Decision Tree, AdaBoost, and MLP

(0.89 for AdaBoost and 0.87 for LightGBM) compared to other tree-based models, suggesting potential challenges with certain data aspects or a need for further tuning to enhance their performance. KNN model proved its effectiveness by achieving low error values (MAE = 13.22, MSE = 289.53, RMSE = 17.02) and a high R-Square of 0.98, successfully capturing the underlying data patterns. Notably, the SVR model underperformed, with high error values (MAE = 119.09, MSE = 17001.72, RMSE = 130.39) and a low R-Square of 0.26, potentially due to its inability to handle specific data characteristics effectively or a need for advanced kernel functions or parameter tuning. The MLP also exhibited poor performance, with extremely high error values (MAE = 104.0320, MSE = 14865.83, RMSE = 121.93) and an R-Square of 0.35, indicating that the model may have overfitted or underfitted the data, failing to capture the underlying relationships accurately.

Strong performance was provided by the ensemble models, which integrate the strengths of various individual models. With an R-Square of 0.99 (MAE = 8.89, MSE = 148.33, RMSE12.17), Ensemble 1, which incorporates Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting, and XGBoost, demonstrated a strong performance. XGBoost, Random Forest, Linear Regression, and LightGBM in Ensemble 2 produced an R-Square of 0.97 (MAE = 16.58, MSE = 594.87, RMSE = 24.39). With an R-Square of 0.99 (MAE = 7.2881, MSE = 107.47, RMSE = 10.37), Ensemble 3, which incorporates Ridge, Lasso, Decision Tree, AdaBoost, and MLP, also performed well. The use of ensemble models illustrated the possibility for enhanced performance by utilizing the strengths of several models, but it also brought attention to the higher computational demands connected with these approaches.

3.4 Comparative Evaluation of the Applied Models

These results highlight the importance of considering the individual attributes of the problem, the type of data, and the assumptions of models when selecting the most suitable approach. Additionally, evaluating models on unseen test data and conducting appropriate model selection techniques can further enhance the reliability and generalization of the chosen model for predicting the WQI. The actual and predicted values of the WQI for all the evaluated models are shown in Fig. 7. Understanding the plot of actual versus predicted values provides a clear assessment of a model's performance. Usually, in this plot, the x-axis displays actual values, while the predicted values are displayed by the y-axis, where each data point represents a pair of ground truth and model predictions. All points falling on the diagonal line (y = x), indicate perfect prediction accuracy as the predictions completely match the actual values. Nevertheless, the variations from this diagonal line expose the errors of the model while points scattered around it show the magnitude and direction of these errors. Points above the diagonal signify that predictions are consistently higher than actual values, while points below indicate underestimation. Analyzing this scatter distribution provides an understanding of the overall working of the model and highlights sections where refinement or adjustment may be necessary to improve predictive accuracy.

4 Discussions

The study's outcomes demonstrate how well ML techniques work to predict the WQI, a crucial metric for evaluating water quality and informing water resource management decisions. The primary objective of this study was to conduct a comprehensive benchmarking of various machine learning models, encompassing linear, tree-based, and ensemble methods. This approach facilitated the capture of a broad spectrum of modeling capabilities and performance characteristics. By including both high-performing models and those with comparatively lower accuracy, such as LightGBM and AdaBoost, a well-rounded evaluation of techniques suitable for predicting WQI in hyper-arid regions was achieved.

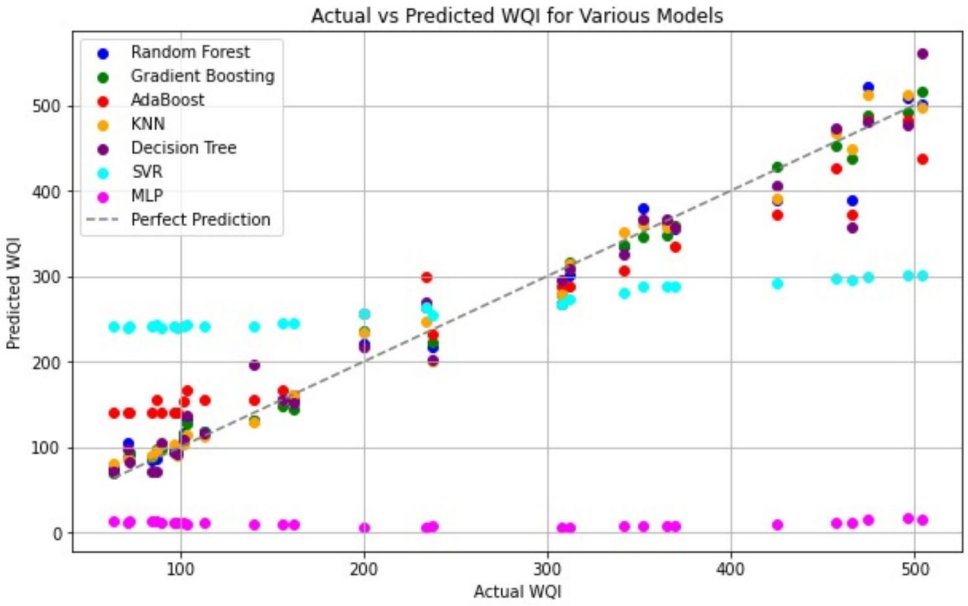


Fig. 7 Figure 7 Actual vs Predicted WQI for various models

The comparative analysis presented in Table 3 highlights the effectiveness of various ML techniques in predicting water quality indices across different studies. Across multiple investigations, different ML algorithms such as DT, RF, SVM, Neural Networks, and GB were employed to assess water quality parameters. Notably, several studies reported high prediction accuracies.

The evaluation of various ML algorithms revealed their distinct strengths, weaknesses, and suitability for this task, providing valuable insights for future applications. Linear models, such as LR and Lasso Regression, demonstrated exceptional performance, accurately capturing the linear correlations between the WQI and water quality parameters. Their simplicity, interpretability, and negligible error rates make them compelling choices for practical applications where model transparency and ease of interpretation are of maximum importance. These models could be specifically useful in situations where stakeholders require clear explanations of the relationships among input variables and the predicted WQI, enabling informed decision-making and communication.

Tree-based models, specifically ExT and GB excelled in capturing non-linear relationships and intricate patterns within the data. Their ability to handle complex interactions and potential non-linearities in water quality parameters makes them well-suited for scenarios where the underlying relationships are not necessarily linear. These models could be invaluable in regions with diverse environmental conditions, varied pollution sources, or complex hydrogeological dynamics that may introduce non-linear effects on water quality.

The deliberate inclusion of underperforming models provided valuable insights. Despite the suboptimal performance of models like LightGBM and AdaBoost, their assessment was essential for understanding the limitations of certain algorithms under extreme environmental conditions. Such insights are particularly relevant for future research, where model optimization may enhance their effectiveness through parameter tuning or the utilization of larger, more diverse datasets. This complete evaluation not only identifies the most suitable models for water quality prediction but also highlights areas where specific algorithms may require further refinement to address challenges unique to hyper-arid environments. Several factors may have contributed to their suboptimal results. First, these models can be sensitive to data distribution; given our dataset's size of 145 samples and four features, the limited data may not suffice for capturing complex patterns effectively. Furthermore, while ensemble methods aim to enhance performance, increased complexity can introduce noise in limited datasets, particularly affecting models like AdaBoost. Finally, LightGBM is optimized for larger datasets and may struggle with smaller samples due to its reliance on gradient-based sampling.

The KNN algorithm also proved strong and effective in capturing underlying data patterns, making it a viable option for water quality prediction tasks. Its non-parametric nature and ability to adapt to local data distributions could be beneficial in heterogeneous environments or when dealing with localized water quality issues. However, the study also revealed limitations in certain models. AdaBoost and Light GBM exhibited relatively higher error rates and lower R-Square scores, indicating potential challenges with specific data characteristics or a need for further model tuning.

Table 3 Comparison of using different ML techniques for predicting water quality

References	Techniques	Best model	Prediction Index	Results
(Radhakrishnan and Pillai 2020)	SVM, Naïve Bayes, DT	DT Algorithm	Weighted Arithmetic Water Quality Index (WAWQI)	Accuracy = 98.50%
(Jain et al. 2021)	SVM, Naïve Bayes, DT	RF Algorithm	WQI	Accuracy = 92.127%
(Hmoud Al-Adhaileh and Waselallah Alsaade 2021)	RF, SVM, KNN	Feed-forward neural network (FFNN), KNN, Neuro-Fuzzy Inference (ANFIS)	WQI, WQC	Accuracy (FFNN) = 100% Accuracy (ANFIS) = 96.17%
(Malek et al. 2022)	ANFIS, KNN, FFNN	GB	WQC	Accuracy = 94.90%
(Khan et al. 2022)	DT, Naive Bayes, GB, KNN, ANN, RF, SVM	GB Classifier	WQI, Water Quality Status (WQS),	Accuracy (PCR) = 95% Accuracy (GBoost) = 100%
(Aldhyani et al. 2020)	Principal Component Regression (PCR), GB Classifier (GBoost)	SVM for WQC and NARNET for (WQI)	WQC	Accuracy (SVM) = 97.01% R2 (NARNET) = 96.17
(Khoi et al. 2022)	Neural Autoregressive Network (NARNET), LSTM, SVM, Naive Bayes & KNN	(XGBoost)	WQI	R2 = 0.989 and RMSE = 0.107
This study	Ridge Model, Lasso Model, RF Model, ExT Model, AB Model, XGB Model, GB) Model, LightGBM Model, LR Model, KNN Model, R Model, SVR Model, DT Model and MLP Regressor Model	LR and Lasso Regression	WQI	(MAE = 0.00, MSE = 0.00, RMSE = 0.00, R-Square = 1.00)

Ensemble techniques have been employed to enhance the performance of our models for water quality prediction, requiring further additional feature engineering and hyperparameter optimization.

The SVR model underperformed, likely due to issues with kernel function selection and parameter tuning. While SVR has been successful in various applications, its performance in this study suggests that further investigation into appropriate kernel functions and regularization techniques may be necessary to leverage its full potential for water quality prediction tasks. The MLP model suffered from severe overfitting or underfitting, which could be attributed to suboptimal model architecture, training strategies, or a lack of appropriate regularization techniques. Addressing these issues through techniques such as dropout, early stopping, or advanced optimization algorithms may improve the performance of neural network models for water quality prediction.

The findings highlight the importance of careful model selection and evaluation for water quality monitoring and management applications. While linear models may suffice in certain scenarios, the complexity of water quality dynamics in arid regions like Abu Dhabi, (UAE), may need the use of more advanced non-linear models. Ensemble techniques, combining the strengths of multiple models, further enhanced prediction accuracy and strength. Future studies should focus on incorporating more diverse datasets, capturing a broader range of environmental conditions and water quality parameters. Additionally, exploring advanced ML techniques, such as deep learning architectures and hybrid models, may potentially uncover more intricate patterns and relationships within the data.

Furthermore, integrating domain-specific knowledge and expert insights into the modeling process could improve the interpretability and practical applicability of the models. For instance, incorporating hydrogeological information, pollution sources, and land-use patterns could provide valuable context and enhance the models' ability to generalize to different regions or scenarios. Overall, this study highlights the potential of ML methods in water quality assessment and management, particularly in arid regions facing water scarcity challenges. By monitoring the effectiveness of different algorithms and addressing their constraints, researchers and experts can boost the precision and consistency of tools

for predicting water quality indices. This enables the implementation of practical measures to protect water resources and ensure sustainable water supply for communities.

5 Conclusion

This comprehensive study has demonstrated the efficacy of various ML techniques in predicting the WQI for groundwater samples collected in the hyper-arid region of Abu Dhabi, (UAE). LR and Lasso Regression models showed exceptional accuracy, capturing linear relationships between water quality parameters and WQI with negligible error rates. Ensemble tree-based models like ExT and GB effectively modeled non-linear relationships and achieved high predictive accuracy. In contrast, AdaBoost, LightGBM, SVR, and MLP exhibited higher error rates and lower predictive accuracy, indicating potential areas for further improvement. These findings highlight the potential of ML approaches in water quality assessment and highlight the importance of well-judged model selection, accurate parameter tuning, and careful data preprocessing.

Future research directions could include exploring additional data sources such as remote sensing and geospatial information to enhance model accuracy and strength. Employing advanced ML techniques, including deep learning models and hybrid approaches, as well as integrating domain-specific knowledge, could further improve model performance. Experimenting different neural network architectures, regularization techniques, and optimization algorithms may address issues of overfitting and underfitting observed in some models. The overall significance of optimizing ML models for WQI prediction lies in the potential to revolutionize water quality monitoring and management, leading to more informed decision-making, timely interventions, and improved water resource management practices.

6 Future Recommendations

In light of the findings and insights from this study on predicting WQI values using ML models, several future recommendations emerge. Integrating more physicochemical parameters and environmental variables into the predictive models could improve their accuracy and broaden the scope of water quality assessments beyond conventional indices. Efforts to enhance data sharing and collaboration among stakeholders, including government agencies, research institutions, and water utilities, are crucial. Such collaborations could facilitate the development of comprehensive, data-driven approaches to water quality management and monitoring, particularly in Abu Dhabi and similar arid regions. This collaborative approach would support more effective and timely interventions to ensure safe and clean water supplies.

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Data Availability Data is available at https://doi.org/10.1016/j.gsd.2021.100611.

Declarations

Conflict of interest The authors declare that they have no competing interests. There are no financial, personal, or professional relationships that could influence the research presented in this article.

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